SERVICE - ORIENTED SOFTWARE ENGINEERING

DT0203

Open Data Formats

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Why Formats Matter to Open Data?

Definition of Open v2.0 from Open Definition

« Open means anyone can freely access, use, modify, and share for any purpose - subject, at most, to requirements that preserve provenance and openness »

legally open

applying an appropriate open license

technically open

using electronic formats that are machine readable and non-proprietary

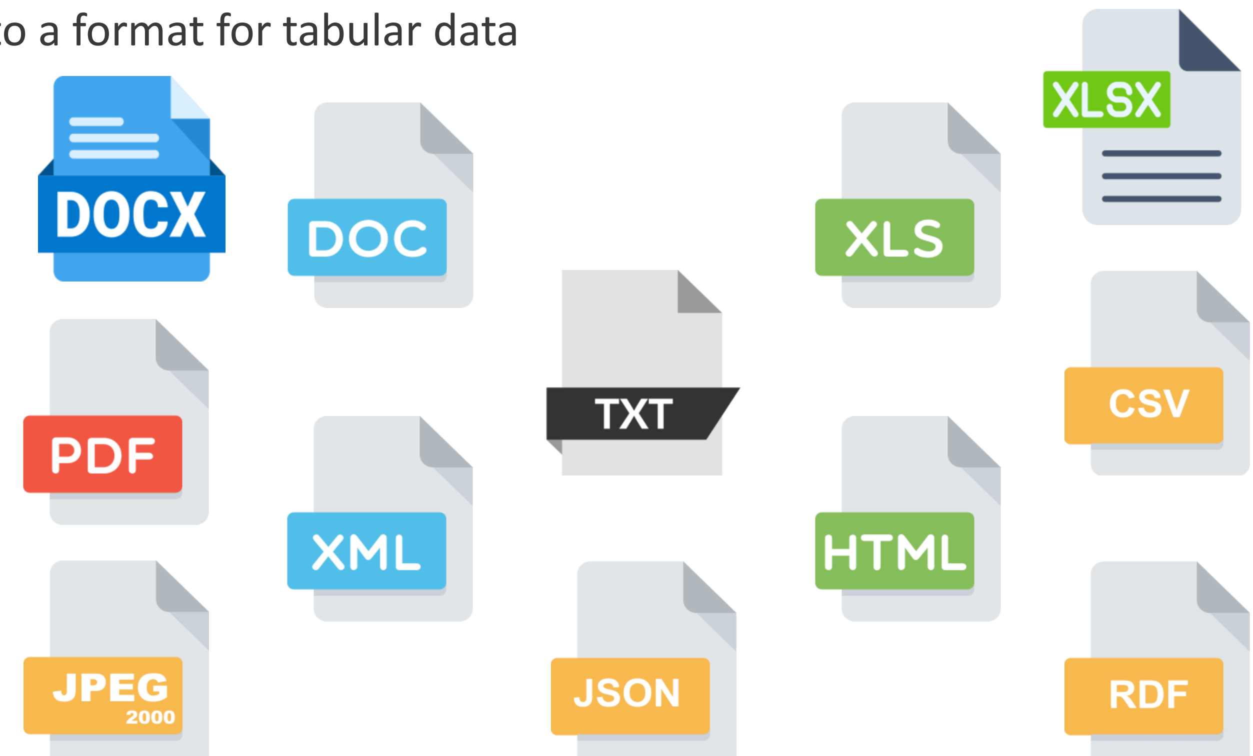
Why Formats Matter to Open Data?

- An open dataset should be machine-readable (as well as human-readable), i.e., it should be in a format that can be **easily** read, written, parsed and displayed by a computer (as well as human)
- Different format for different media types
  - e.g., image, text , . . .
- Different format for different data structures
  - e.g., tabular, hierarchical, . . .

What Format Matter to Open Data?

Maximize the access, use, reuse, and share of the data

Appropriately expressing what the data represents, e.g., format for geographical data may be different to a format for tabular data



Common Data Structures

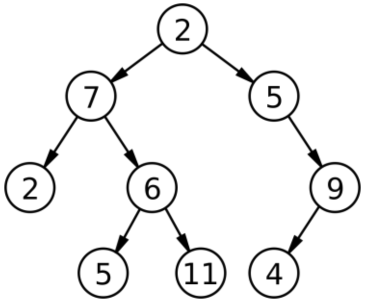
Tabular (CSV)

- Comma-separated values
- data is organized into rows and columns

Professions	% of People	% of Males	% of Females
Medical	11	60	40
Engineering	18	30	70
Law	24	45	55
Teaching	21	80	20
Banking	16	35	65
Management	10	44	56

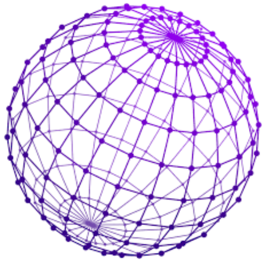
Hierarchical

- hierarchical structured data allows relationships between data by following a vertical hierarchical structure, like tree



Network

- network structured data allows relationships between any combination of elements in any direction



File Format Overview

Text document

- documents in formats like DOC or PDF are human-readable, but they may not be "straightforwardly machine-readable" because of formatting metadata, different font types, images, etc.

- e.g., DOC or PDF documents containing tables of data are definitely digital but a computer may struggle to access the tabular information

- generally, it is recommended not to exhibit data in DOC (proprietary format) or PDF (although on 2008, from v1.7, PDF has become non-proprietary format, now PDF 2.0)

http://opendatahandbook.org/guide/en/appendices/file-formats

File Format Overview

Text document

Table 1. Expenditures by local category, all households, United States, 2014

	Mean spending, all households	**Share of local purchases**
Housing	$15,895	60%
Utilities, fuel, and public services	$3,921	15%
Food away from home	$2,787	11%
Vehicle maintenance	$836	3%
Medical services	$790	3%
Personal care products and services	$645	2%
Entertainment fees and admission	$636	2%
Public transportation	$581	2%
Personal household services	$366	1%

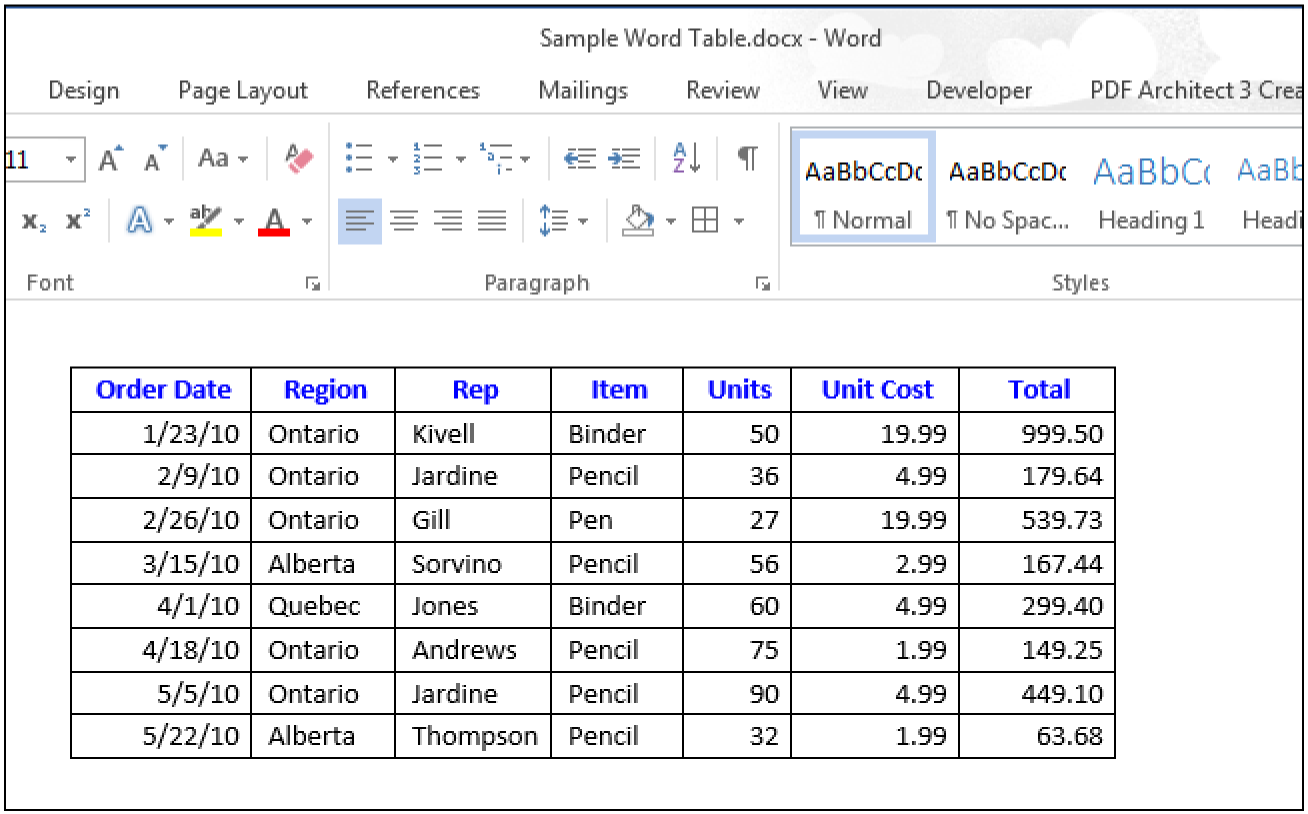
Note: Housing expenses exclude property taxes, which are included in state and local taxes.

Source: Brookings analysis of Bureau of Labor Statistics 2014 Consumer Expenditure Survey.

http://opendatahandbook.org/guide/en/appendices/file-formats

File Format Overview

Text document



Order Date	Region	Rep	Item	Units	Unit Cost	Total
1/23/10	Ontario	Kivell	Binder	50	19.99	999.50
2/9/10	Ontario	Jardine	Pencil	36	4.99	179.64
2/26/10	Ontario	Gill	Pen	27	19.99	539.73
3/15/10	Alberta	Sorvino	Pencil	56	2.99	167.44
4/1/10	Quebec	Jones	Binder	60	4.99	299.40
4/18/10	Ontario	Andrews	Pencil	75	1.99	149.25
5/5/10	Ontario	Jardine	Pencil	90	4.99	449.10
5/22/10	Alberta	Thompson	Pencil	32	1.99	63.68

http://opendatahandbook.org/guide/en/appendices/file-formats

File Format Overview

Plain text

- documents in formats like TXT are human- and ”easily machine-readable”

- differently from text documents, plain text documents exclude structural metadata from inside the document hence, developers simply need to create a parser that can interpret each document as it appears.

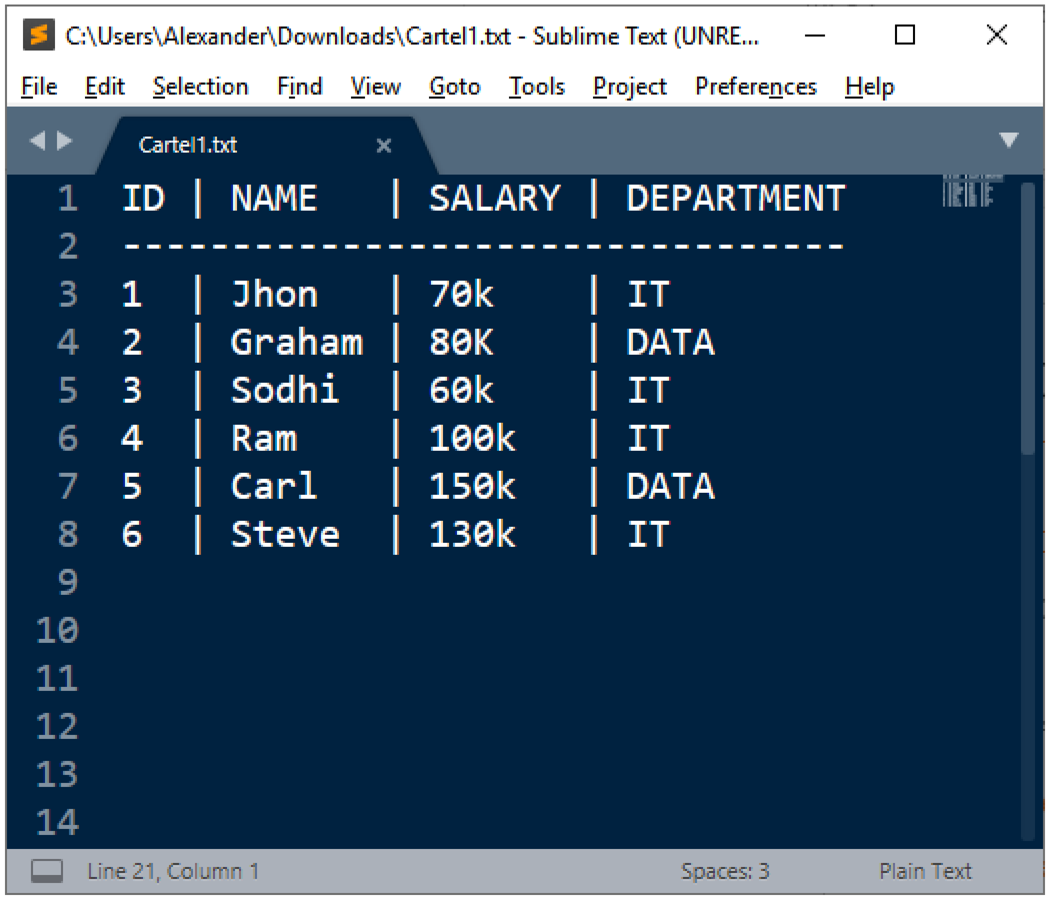
- parsers, however, may have compatibility issue between operating systems, e.g., Windows, Mac OS X and Unix have their own way of telling the computer that they have reached the end of line

http://opendatahandbook.org/guide/en/appendices/file-formats

File Format Overview

TXT

Plain text



http://opendatahandbook.org/guide/en/appendices/file-formats

File Format Overview

Scanned image

- probably the least suitable form for most data
- JPEG-2000 can be enriched with documentation of what the picture shows
- this format may be relevant for displaying as image those data that are not born electronically, e.g., old church records and archival material
- picture is better than nothing !

http://opendatahandbook.org/guide/en/appendices/file-formats

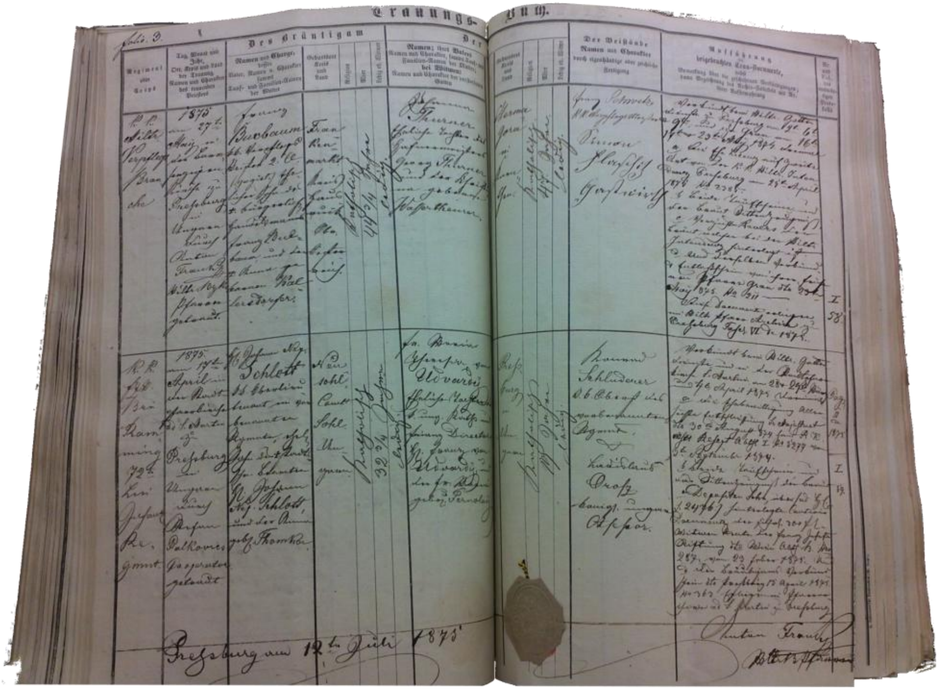
File Format Overview

Scanned image

e.g., military and church registers

include general information like:

- detailed personnel data
- general information about baptisms, marriages, and death
- . . .



http://opendatahandbook.org/guide/en/appendices/file-formats

File Format Overview

Hypertext Markup Language (HTML)

- HTML web pages containing tables of data are human-readable, but they may not be machine-readable, depending on how the data is encoded
- HTML tables are good as starting point in the display of data but could be preferable to have data in a form easier to download and manipulate
- Yahoo developed the Yahoo Query Language (https://en.wikipedia.org/wiki/Yahoo!_Query_Language -- then retired in January 2019) that was able to extracts structured information from websites, but it required to carefully tag the extracted data if you wanted to do more

http://opendatahandbook.org/guide/en/appendices/file-formats

File Format Overview

Hypertext Markup Language (HTML)

C:\Users\Alexander\Downloads\Cartel1.html • - Sublime Text (UNREGIS...

File Edit Selection Find View Goto Tools Project Preferences Help

Cartel1.html

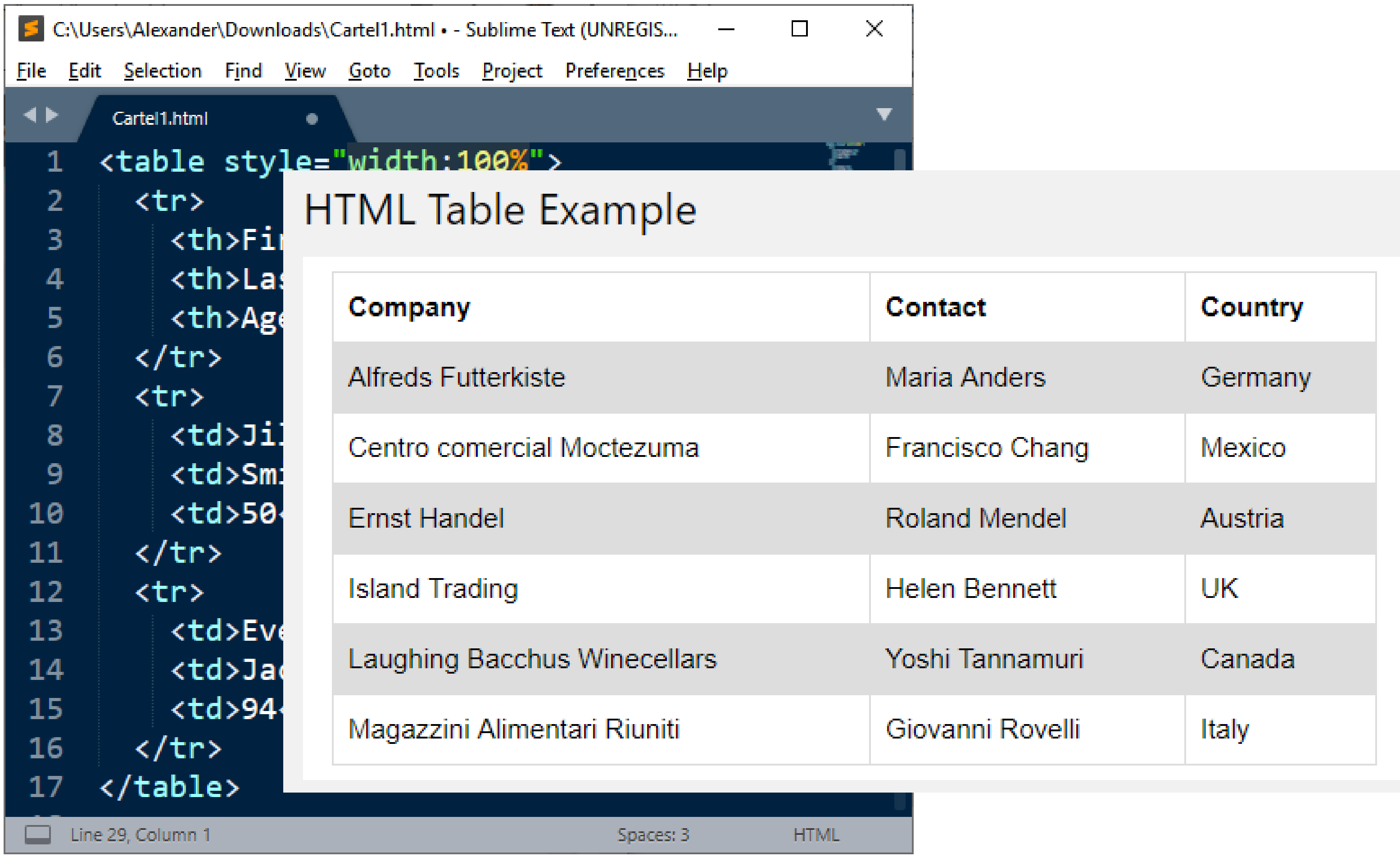
```html
<table style="width:100%">
   <tr>
      <th>Firstname</th>
      <th>Lastname</th>
      <th>Age</th>
   </tr>
   <tr>
      <td>Jill</td>
      <td>Smith</td>
      <td>50</td>
   </tr>
   <tr>
      <td>Eve</td>
      <td>Jackson</td>
      <td>94</td>
   </tr>
</table>
```

Line 29, Column 1 Spaces: 3 HTML

http://opendatahandbook.org/guide/en/appendices/file-formats

File Format Overview

Hypertext Markup Language (HTML)



HTML Table Example

Company	Contact	Country
Alfreds Futterkiste	Maria Anders	Germany
Centro comercial Moctezuma	Francisco Chang	Mexico
Ernst Handel	Roland Mendel	Austria
Island Trading	Helen Bennett	UK
Laughing Bacchus Winecellars	Yoshi Tannamuri	Canada
Magazzini Alimentari Riuniti	Giovanni Rovelli	Italy

http://opendatahandbook.org/guide/en/appendices/file-formats

File Format Overview

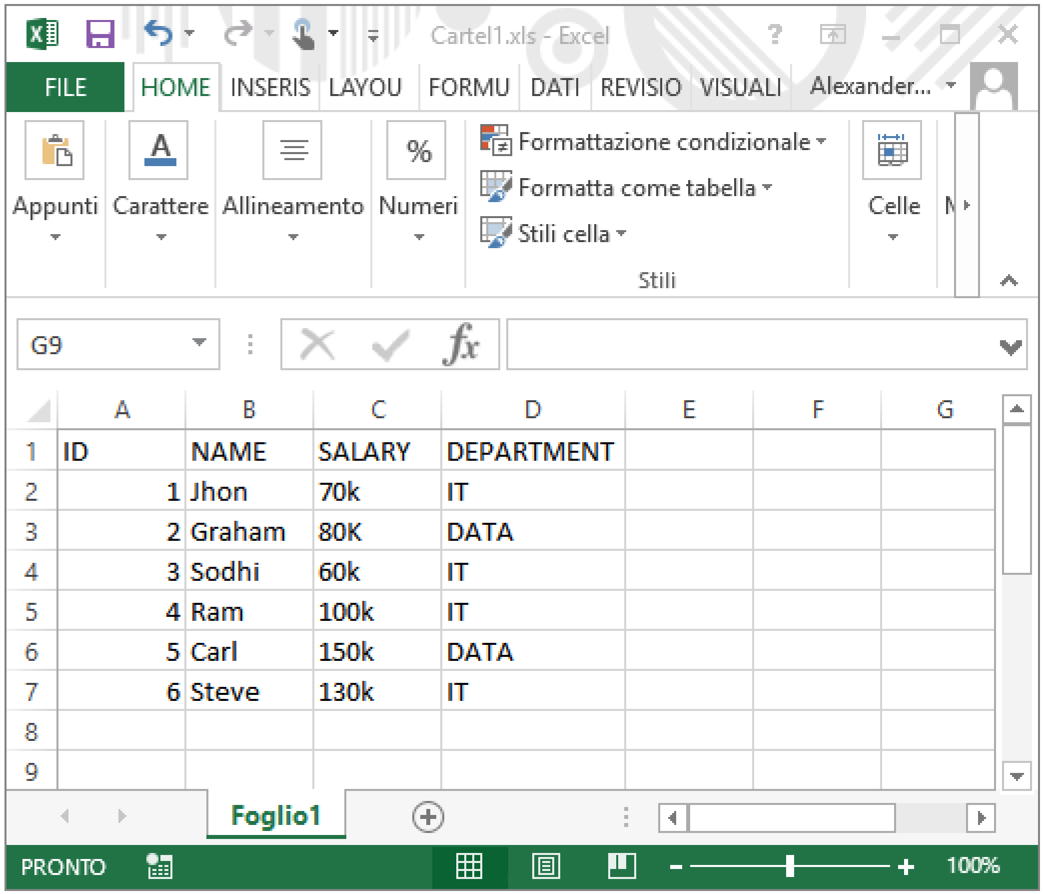
Spreadsheets

- many authorities have information left in the spreadsheet, e.g., XLSX (proprietary format)
- a correct description of what columns mean helps to use these data immediately
- tables in a spreadsheet format would be human- and machine-readable, but in some cases could be difficult to read these formats if macros and formulas are therein. Therefore, it is recommended to document such calculations next to the spreadsheet, to make the spreadsheet more accessible

http://opendatahandbook.org/guide/en/appendices/file-formats

File Format Overview

Spreadsheets



ID	NAME	SALARY	DEPARTMENT
1	Jhon	70k	IT
2	Graham	80K	DATA
3	Sodhi	60k	IT
4	Ram	100k	IT
5	Carl	150k	DATA
6	Steve	130k	IT

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File Format Overview

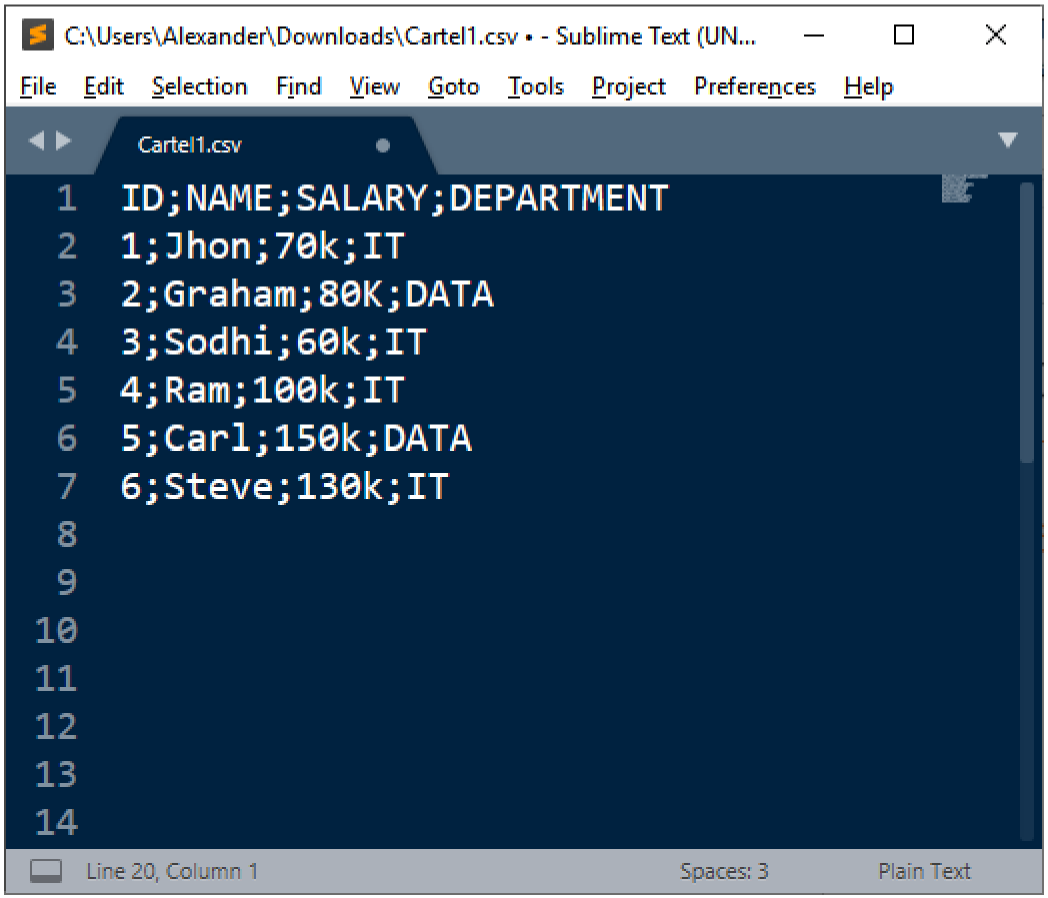
Comma Separated Values (CSV)

- very simple and compact format, suitable to transfer large sets of data with the same structure
- is important for the comma-separated formats that documentation of the individual fields is accurate
- is essential that the structure of the file is respected. A single omission of a field may disturb the reading of all remaining data

http://opendatahandbook.org/guide/en/appendices/file-formats

File Format Overview

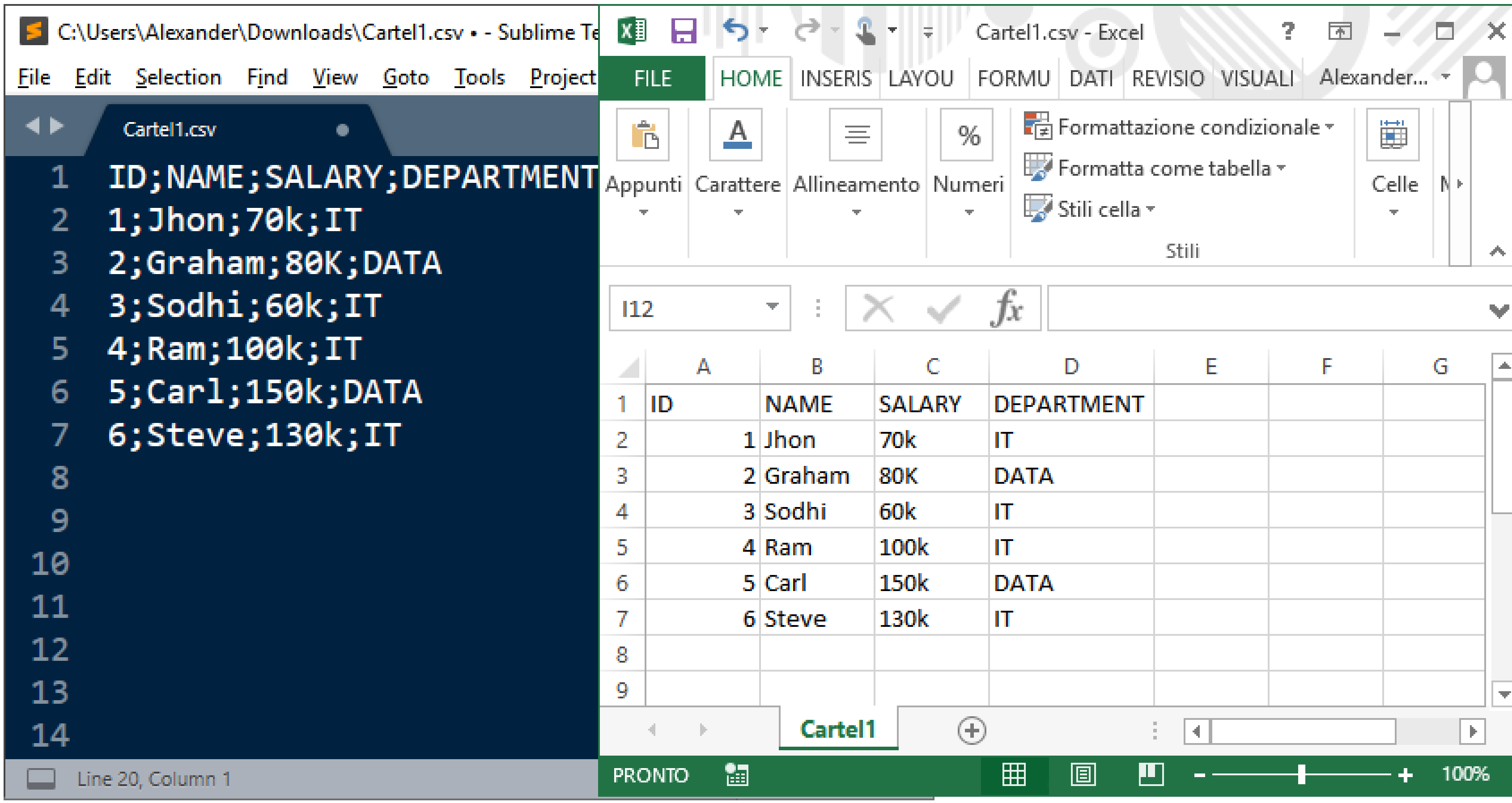
Comma Separated Values (CSV)



http://opendatahandbook.org/guide/en/appendices/file-formats

File Format Overview

Comma Separated Values (CSV)



http://opendatahandbook.org/guide/en/appendices/file-formats

File Format Overview

Extensible Markup Language (XML)

- is a markup language that defines a set of rules for encoding documents in a format that is both human-readable and machine-readable

- is widely used for data exchange

- is widely used for the representation of data structures such as those used in Web Services (also part of the course)

http://opendatahandbook.org/guide/en/appendices/file-formats

File Format Overview

Extensible Markup Language (XML)

```xml
<Books>
    <Book ISBN="0553212419">
        <title>Sherlock Holmes: Complete Novels...
        <author>Sir Arthur Conan Doyle</author>
    </Book>
    <Book ISBN="0743273567">
        <title>The Great Gatsby</title>
        <author>F. Scott Fitzgerald</author>
    </Book>
    <Book ISBN="0684826976">
        <title>Undaunted Courage</title>
        <author>Stephen E. Ambrose</author>
    </Book>
    <Book ISBN="0743203178">
        <title>Nothing Like It In the World</title>
        <author>Stephen E. Ambrose</author>
    </Book>
</Books>
```

http://opendatahandbook.org/guide/en/appendices/file-formats

File Format Overview

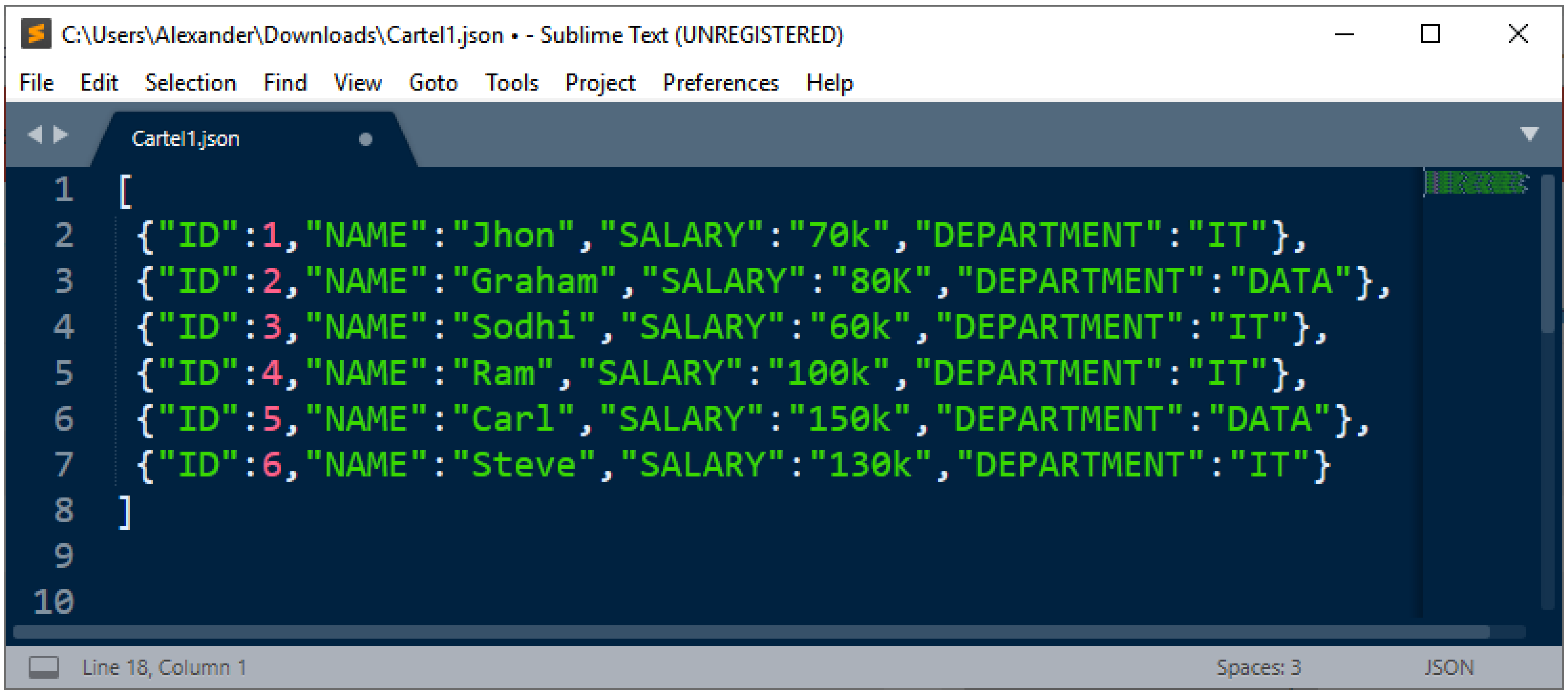
JavaScript Object Notation (JSON)

- is an open-standard file format that uses human-readable text to transmit data structured in attribute-value pairs and array data types
- is very easy to read by any programming language (machine-readable), better than XML
- geoJSON format for representing simple geographical data
- widely used not only for Open Data, but also for Web Services

http://opendatahandbook.org/guide/en/appendices/file-formats

File Format Overview

JavaScript Object Notation (JSON)



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File Format Overview

Resource Description Framework (RDF)

- World Wide Web Consortium (W3C) recommended format
- Is used to represent the relationship between data on the web
- It is a form of a graph-based database
- makes it possible to represent data in a form that makes it easier to combine data from multiple sources. RDF data can be stored in XML and JSON syntax (human- and machine-readable)
- encourages the use of URLs as identifiers, which provides a convenient way to directly interconnect existing open data initiatives on the Web
- is still not widespread, but it has been a trend among Open Government initiatives
- RDF document can be queried with SPARQL query language

http://opendatahandbook.org/guide/en/appendices/file-formats

File Format Overview

Resource Description Framework (RDF)

- Based on the idea of making statements about resources
- An RDF Statement consists 3 components triples (Subject-Predicate-Object)
    - Subject denotes the resource
    - Predicate denotes a distinguishing quality, characteristic or aspect of the resource, and expresses a relationship between the subject and the object.
    - Object are things related to the subject.

  Example: notion "The sky has the color blue" in RDF is the triple: a subject denoting "the sky", a predicate denoting "has the color", and an object denoting "blue"
- mechanism for describing resources is a major component in the W3C's Semantic Web activity
- A collection of RDF statements represents a labeled, directed multi-graph
- RDF is an abstract model with several serialization formats (i.e. file formats), so the particular encoding for resources or triples varies from format to format

https://en.wikipedia.org/wiki/Resource_Description_Framework

RDF - Serialization Syntax: RDF/XML

- RDF/XML is a serialization format for RDF that uses XML syntax to represent RDF triples (subject–predicate–object).
- it's the oldest one so most applications offer support to it
- avoid unnecessary repetition of URIs using namespaces as shortcuts
- most RDF libraries use this format by default
- is not the best one with respect human-readability
- the file cannot be splitted during streaming

```xml
<?xml version="1.0" encoding="utf-8" ?>
<rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
         xmlns:foaf="http://xmlns.com/foaf/0.1/"
         xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#"
         xmlns:ns0="http://dbpedia.org/ontology/"
         xmlns:schema="http://schema.org/"
         xmlns:geo="http://www.w3.org/2003/01/geo/wgs84_pos#">

  <foaf:Person rdf:about="http://dbpedia.org/resource/Bob_Marley">
    <rdfs:label xml:lang="en">Bob Marley</rdfs:label>
    <rdfs:label xml:lang="fr">Bob Marley</rdfs:label>
    <rdfs:seeAlso rdf:resource="http://dbpedia.org/resource/Rastafari"/>
    <ns0:birthPlace>
      <schema:Country rdf:about="http://dbpedia.org/resource/Jamaica">
        <rdfs:label xml:lang="en">Jamaica</rdfs:label>
        <rdfs:label xml:lang="it">Giamaica</rdfs:label>
        <geo:lat rdf:datatype="http://www.w3.org/2001/XMLSchema#float">17.9833</geo:lat>
        <geo:long rdf:datatype="http://www.w3.org/2001/XMLSchema#float">-76.8</geo:long>
        <foaf:homepage rdf:resource="http://jis.gov.jm/"/>
      </schema:Country>
    </ns0:birthPlace>

  </foaf:Person>

</rdf:RDF>
```

RDF - Serialization Syntax: RDF/XML

```xml
<?xml version="1.0" encoding="utf-8" ?>
<rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
        xmlns:foaf="http://xmlns.com/foaf/0.1/"
        xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#"
        xmlns:ns0="http://dbpedia.org/ontology/"
        xmlns:schema="http://schema.org/"
        xmlns:geo="http://www.w3.org/2003/01/geo/wgs84_pos#">

  <foaf:Person rdf:about="http://dbpedia.org/resource/Bob_Marley">
    <rdfs:label xml:lang="en">Bob Marley</rdfs:label>
    <rdfs:label xml:lang="fr">Bob Marley</rdfs:label>
    <rdfs:seeAlso rdf:resource="http://dbpedia.org/resource/Rastafari"/>
    <ns0:birthPlace>
      <schema:Country rdf:about="http://dbpedia.org/resource/Jamaica">
        <rdfs:label xml:lang="en">Jamaica</rdfs:label>
        <rdfs:label xml:lang="it">Giamaica</rdfs:label>
        <geo:lat rdf:datatype="http://www.w3.org/2001/XMLSchema#float">17.9833</geo:lat>
        <geo:long rdf:datatype="http://www.w3.org/2001/XMLSchema#float">-76.8</geo:long>
        <foaf:homepage rdf:resource="http://jis.gov.jm/"/>
      </schema:Country>
    </ns0:birthPlace>

  </foaf:Person>

</rdf:RDF>
```

foaf: person

RDF.XML uses namespaces to avoid repetition

This reduces repetition and makes the document shorter.

RDF - Serialization Syntax: Turtle

- both highly human- and machine-readable
- avoid unnecessary repetition of URIs using prefixes as shortcuts
- allows hierarchical grouping
- preferred format in case of bandwidth issues
- can be streamed in blocks due to the lack of opening and closing lines

```
@prefix dbr:    <http://dbpedia.org/resource/> .
@prefix dbo:    <http://dbpedia.org/ontology/> .
@prefix rdfs:   <http://www.w3.org/2000/01/rdf-schema#> .
@prefix foaf:   <http://xmlns.com/foaf/0.1/> .
@prefix geo:    <http://www.w3.org/2003/01/geo/wgs84_pos#> .
@prefix xsd:    <http://www.w3.org/2001/XMLSchema#> .
@prefix schema: <http://schema.org/> .

dbr:Bob_Marley
    a foaf:Person ;
    rdfs:label "Bob Marley"@en ;
    rdfs:label "Bob Marley"@fr ;
    rdfs:seeAlso dbr:Rastafari ;
    dbo:birthPlace dbr:Jamaica .

dbr:Jamaica
    a schema:Country ;
    rdfs:label "Jamaica"@en ;
    rdfs:label "Giamaica"@it ;
    geo:lat "17.9833"^^xsd:float ;
    geo:long "-76.8"^^xsd:float ;
    foaf:homepage <http://jis.gov.jm/> .
```

RDF - Serialization Syntax: N-Triples

- effective when verbosity and readability are not a concern
- highly efficient compression ratios
- allows streaming line by line
- not properly human-readability
- does not allow abbreviations

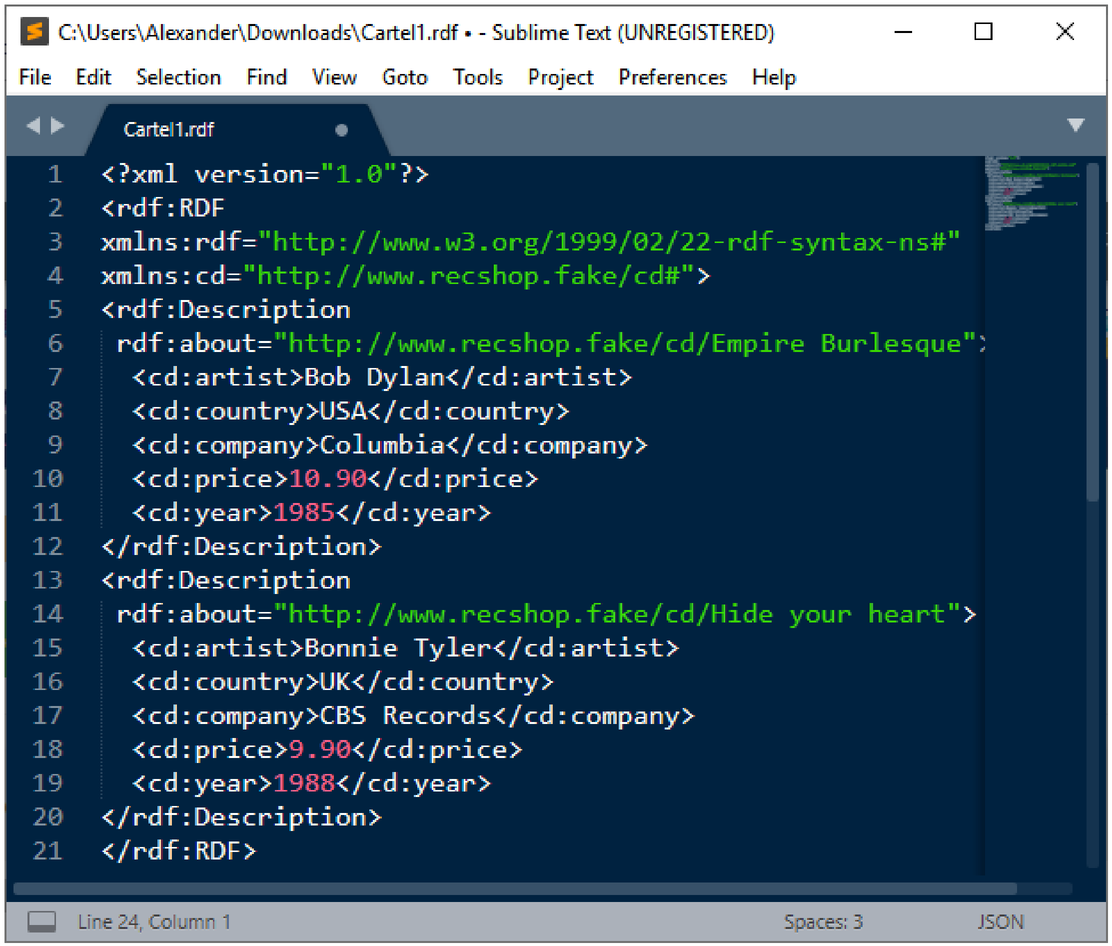
N-Triple Notation

- URIs/IRIs in angle brackets
- Literals in question marks
- Triple ends with a period

```
<http://dbpedia.org/resource/Bob_Marley> <http://www.w3.org/1999/02/22-rdf-syntax-ns#type> <http://xmlns.com/foaf/0.1/Person> .
<http://dbpedia.org/resource/Bob_Marley> <http://www.w3.org/2000/01/rdf-schema#label> "Bob Marley"@en .
<http://dbpedia.org/resource/Bob_Marley> <http://www.w3.org/2000/01/rdf-schema#label> "Bob Marley"@fr .
<http://dbpedia.org/resource/Bob_Marley> <http://www.w3.org/2000/01/rdf-schema#seeAlso> <http://dbpedia.org/resource/Rastafari> .
<http://dbpedia.org/resource/Bob_Marley> <http://dbpedia.org/ontology/birthPlace> <http://dbpedia.org/resource/Jamaica> .
<http://dbpedia.org/resource/Jamaica> <http://www.w3.org/1999/02/22-rdf-syntax-ns#type> <http://schema.org/Country> .
<http://dbpedia.org/resource/Jamaica> <http://www.w3.org/2000/01/rdf-schema#label> "Jamaica"@en .
<http://dbpedia.org/resource/Jamaica> <http://www.w3.org/2000/01/rdf-schema#label> "Giamaica"@it .
<http://dbpedia.org/resource/Jamaica> <http://www.w3.org/2003/01/geo/wgs84_pos#lat> "17.9833"^^<http://www.w3.org/2001/XMLSchema#float> .
<http://dbpedia.org/resource/Jamaica> <http://www.w3.org/2003/01/geo/wgs84_pos#long> "-76.8"^^<http://www.w3.org/2001/XMLSchema#float> .
<http://dbpedia.org/resource/Jamaica> <http://xmlns.com/foaf/0.1/homepage> <http://jis.gov.jm/> .
```

File Format Overview

Resource Description Framework (RDF)




```xml
<?xml version="1.0"?>
<rdf:RDF
xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
xmlns:cd="http://www.recshop.fake/cd#">
<rdf:Description
 rdf:about="http://www.recshop.fake/cd/Empire Burlesque">
  <cd:artist>Bob Dylan</cd:artist>
  <cd:country>USA</cd:country>
  <cd:company>Columbia</cd:company>
  <cd:price>10.90</cd:price>
  <cd:year>1985</cd:year>
</rdf:Description>
<rdf:Description
 rdf:about="http://www.recshop.fake/cd/Hide your heart">
  <cd:artist>Bonnie Tyler</cd:artist>
  <cd:country>UK</cd:country>
  <cd:company>CBS Records</cd:company>
  <cd:price>9.90</cd:price>
  <cd:year>1988</cd:year>
</rdf:Description>
</rdf:RDF>
```

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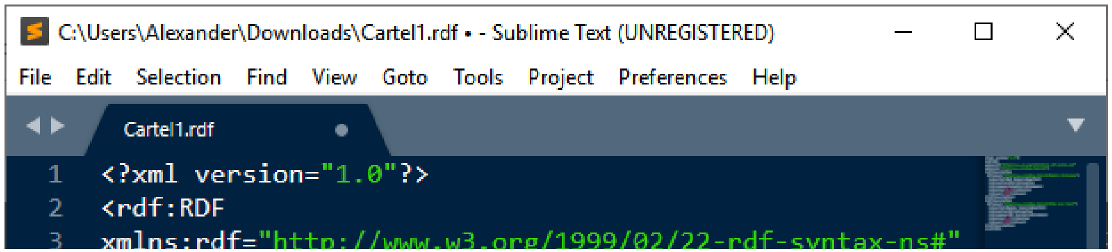
File Format Overview

Resource Description Framework (RDF)



Triples of the Data Model

Number	Subject	Predicate	Object
1	http://www.recshop.fake/cd/Empire Burlesque	http://www.recshop.fake/cd#artist	"Bob Dylan"
2	http://www.recshop.fake/cd/Empire Burlesque	http://www.recshop.fake/cd#country	"USA"
3	http://www.recshop.fake/cd/Empire Burlesque	http://www.recshop.fake/cd#company	"Columbia"
4	http://www.recshop.fake/cd/Empire Burlesque	http://www.recshop.fake/cd#price	"10.90"
5	http://www.recshop.fake/cd/Empire Burlesque	http://www.recshop.fake/cd#year	"1985"
6	http://www.recshop.fake/cd/Hide your heart	http://www.recshop.fake/cd#artist	"Bonnie Tyler"
7	http://www.recshop.fake/cd/Hide your heart	http://www.recshop.fake/cd#country	"UK"
8	http://www.recshop.fake/cd/Hide your heart	http://www.recshop.fake/cd#company	"CBS Records"
9	http://www.recshop.fake/cd/Hide your heart	http://www.recshop.fake/cd#price	"9.90"
10	http://www.recshop.fake/cd/Hide your heart	http://www.recshop.fake/cd#year	"1988"

http://opendatahandbook.org/guide/en/appendices/file-formats

Open Formats

open format for data

- XML (eXtensible Markup Language)
- N-Triples (serialization format for RDF)
- Notation3 (serialization format for RDF)
- Turtle (serialization format for RDF)
- JSON-LD (serialization format for RDF)
- JSON (JavaScript Object Notation)
- CSV (Comma Separated Values)

popular open formats for geographic data

- Shapefile
- KML
- GeoJSON
- GML (Geography Markup Language)
- GeoPackage

open formats for documents

- ODF (Open Document Format)
- PDF
- Akoma Ntoso

The Choice of an Open Data Publisher

Not all data are suitable for download because

- too large - should the data be split into smaller datasets?
- too regularly updated - how often should the data be updated?
- too complex to expose as static files - how changes will affect the previous releases?

Data are exposed through web services and they are accessible through an application programming interface (API)

- e.g., https://hello.irail.be/api/1-0/
- try with curl https://irail.be/stations/NMBS?q=Brussels

References

- http://opendatahandbook.org/guide/en/appendices/file-formats
- https://www.w3.org/2009/Talks/0615-qbe/